Chapter 5

Eigenvalues and Eigenvectors

5.1 Introduction

Eigenvalues and eigenvectors are simply defined. A vector $\mathbf{x}$ is an eigenvector of matrix A with eigenvalue λ if $\mathbf{x} \neq \mathbf{0}$ and

$$A\mathbf{x} = \lambda\mathbf{x}. \tag{5.1}$$

This says that multiplying the eigenvector $\mathbf{x}$ by the matrix A has the same effect as multiplying the vector by the number λ, geometrically this makes the vector longer or shorter, and if $\lambda < 0$ makes the vector point in exactly the opposite direction, but cannot otherwise change the direction of the vector. Note that if A is an $n \times m$ matrix and $\mathbf{x}$ is an m vector then $A\mathbf{x}$ is an n vector and $\lambda\mathbf{x}$ is an m vector, so the only matrices that can have eigenvectors and eigenvalues are square $n \times n$ matrices.

It is not immediately obvious why economists might be interested in eigenvalues and eigenvectors. In fact they turn out to be very important in several contexts. One is dealing with quadratic forms which are what quadratic functions turn into when they grow up. You will meet these in mathematics for microeconomics. They also matter in econometrics and finance (they underpin the Capital Asset Pricing Model). The other area where eigenvalues and eigenvectors are important is economic dynamics, studied with difference and differential equations. This is currently extremely trendy. Hence the need to know something about eigenvalues and eigenvectors.

5.2 Finding Eigenvalues: the Characteristic Polynomial

Equation 5.1 can be rewritten as

$$(A - \Lambda)\,\mathbf{x} = \mathbf{0} \tag{5.2}$$

where Λ is a diagonal matrix, that is all its off diagonal terms are 0, and all its diagonal terms are λ. As $\mathbf{x} \neq 0$ equation 5.2 and the result that $\det B = 0$ if and only if there is a non zero vector $\mathbf{x}$ such that $B\mathbf{x} = \mathbf{0}$ implies that

$$\det(A - \Lambda) = 0$$

or writing this out in full

$$\begin{vmatrix} A_{11} - \lambda & A_{12} & . & . & A_{1n} \\ A_{21} & A_{22} - \lambda & . & . & A_{2n} \\ . & . & . & . & . \\ . & . & . & . & . \\ A_{n1} & A_{n2} & & & A_{nn} - \lambda \end{vmatrix} = 0. \tag{5.3}$$

Thus all you have to do to find the eigenvalues of the matrix is solve this equation. Sadly that is often not an easy task. Firstly you have to calculate the determinant; this is a polynomial of degree n called the **characteristic polynomial** $\det(A - \Lambda)$. Then you have to solve the **characteristic equation** $\det(A - \Lambda) = 0$. The fundamental theorem of algebra (see Background Notes on polynomials) tells you that for any polynomial $p(\lambda)$ in λ of degree n there are n numbers, called the roots of the polynomial with the property that

$$p(\lambda) = (\lambda_1 - \lambda)(\lambda_2 - \lambda)\ldots.(\lambda_n - \lambda). \tag{5.4}$$

The number λ_i are called the roots of the polynomial; they are the solutions of the equation $p(\lambda) = 0$, and the eigenvalues of the matrix A. One fact that follows straight from the fact that the determinant in equation 7.11 is equal to the polynomial in equation 5.4 and thinking about what they are when $\lambda = 0$, gives

Proposition 36 *If A is a square matrix the determinant of A is the product of the eigenvalues of A, that is*

$$\det A = \lambda_1\lambda_2\ldots.\lambda_n.$$

5.3 Some Real Matrices Have Complex Eigenvalues and Eigenvectors

Even if the matrix A is formed from real numbers the eigenvalues can be complex. For example the matrix

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

has a characteristic equation $\lambda^2 + 1 = 0$ so has no real roots. In fact if x_1 and x_2 are real numbers

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_2 \\ -x_1 \end{pmatrix}$$

and it is easy to check that the two vectors

$$(x_1, x_2)\begin{pmatrix} x_2 \\ -x_1 \end{pmatrix} = 0$$

so the vectors $\mathbf{x}$ and $A\mathbf{x}$ area orthogonal. (at 90°) to each other, so there is no way that $A\mathbf{x}$ can be a scalar multiple $\lambda\mathbf{x}$ of $\mathbf{x}$ if $\mathbf{x}$ is real. The one general thing that can be said about the eigenvalues of real matrices is that the fundamental theorem of algebra (Background Notes on polynomials) implies that any complex eigenvalues come in pairs of complex conjugates.

5.4 When Finding the Eigenvalues of a Matrix is Easy

5.4.1 The 2×2 Case

In this case the characteristic equation of the matrix is

$$\begin{aligned} \left| \begin{array}{cc} A_{11}-\lambda & A_{12} \\ A_{21} & A_{22}-\lambda \end{array} \right| &= \lambda^2 - \lambda\left(A_{11}+A_{22}\right) + A_{11}A_{22} - A_{12}A_{21} \\ &= \lambda^2 - \lambda\left(trA\right) + \det A \end{aligned}$$

where trA is the trace of A, that is the sum of the diagonal terms, and $\det A$ is the determinant of A. There is of course a formula for the solutions of this equation. For example if

$$A = \left(\begin{array}{cc} 4 & 1 \\ -2 & 1 \end{array} \right)$$

The characteristic equation is

$$\left| \begin{array}{cc} 4-\lambda & 1 \\ -2 & 1-\lambda \end{array} \right| = \lambda^2 - \lambda\left(trA\right) + \det A = \lambda^2 - 5\lambda + 6 = \left(\lambda-2\right)\left(\lambda-3\right)$$

so the eigenvalues are 2 and 3. The eigenvector corresponding to the eigenvalue 2 can be found by solving the equation

$$\left(\begin{array}{cc} 4 & 1 \\ -2 & 1 \end{array} \right) \left(\begin{array}{c} x_1 \\ x_2 \end{array} \right) = 2 \left(\begin{array}{c} x_1 \\ x_2 \end{array} \right)$$

or equivalently

$$\left(\begin{array}{cc} 4-2 & 1 \\ -2 & 1-2 \end{array} \right) \left(\begin{array}{c} x_1 \\ x_2 \end{array} \right) = \left(\begin{array}{cc} 2 & 1 \\ -2 & -1 \end{array} \right) \left(\begin{array}{c} x_1 \\ x_2 \end{array} \right) = \left(\begin{array}{c} 2x_1 + x_2 \\ -2x_1 - x_2 \end{array} \right) = 0$$

so any x_1 and x_2 with $x_2 = -2x_1$ is a solution; $x_1 = 1$, $x_2 = -2$, is the simplest but any vector of the form

$$\beta \left(\begin{array}{c} 1 \\ -2 \end{array} \right)$$

where $\beta \neq 0$ is an eigenvector. This is a general point about eigenvectors, if $\mathbf{x}$ is an eigenvector of matrix A so is $\beta\mathbf{x}$.

A similar argument for the eigenvector corresponding to the eigenvalue 3, says that any vector of the form

$$\beta \left(\begin{array}{c} 1 \\ -1 \end{array} \right)$$

is an eigenvector.

5.4.2 Diagonal, Upper and Lower Triangular Matrices

I showed in chapter 4 on determinants the determinant of a diagonal or upper or lower triangular matrix is the product of the diagonal terms. If A is a diagonal or upper or lower triangular matrix so also is $A-\Lambda$, so the characteristic equation of A

$$(A_{11} - \lambda)\,(A_{22} - \lambda)\,(A_{33} - \lambda)\ldots.\,(A_{nn} - \lambda) = 0 \tag{5.5}$$

so the solutions are the diagonal terms A_{11}, A_{22}, $A_{33}....A_{nn}$. In these cases the eigenvalues are the diagonal terms of the matrix.

5.5 Diagonalizing a Matrix

There are some useful results that I state without proof.

Proposition 37 *If all the eigenvalues of A are all different then there is a matrix M with an inverse M^{-1} with the property that*

$$A = M^{-1}DM$$

where D is a diagonal matrix, whose diagonal terms are the eigenvalues of A.

There are stronger statements to be made about the eigenvalues and eigenvectors of symmetric matrices (the matrix A is symmetric if $A' = A$, so $A_{ij} = A_{ji}$ for all i and j.

Proposition 38 *If the matrix A is symmetric, and all its components are real numbers, then all its eigenvalues are real and the eigenvectors corresponding to different eigenvalues are orthogonal., that is if $\mathbf{x}_1$ and $\mathbf{x}_2$ are eigenvectors of A with eigenvalues λ_1 and λ_2, and $\lambda_1 \neq \lambda_2$ then*

$$\mathbf{x}_1'\mathbf{x}_2 = 0.$$

There is then a fairly natural definition

Definition 39 *The matrix M is* ***orthogonal.*** *if its transpose is also its inverse, so*

$$MM' = M'M = I.$$

Another way of saying this is that any two different columns of M are orthogonal. The big result is then

Proposition 40 *If the matrix A is symmetric and all its components are real numbers, then all its eigenvalues are real and there is an orthogonal. matrix M such that*

$$A = M'DM$$

where D is a diagonal matrix whose diagonal terms are the eigenvalues of I.

For some purposes, notably the solution of difference and differential equations we want to know about A^n. If A is diagonalizable, that is it can be written in the form

$$A = M^{-1}DM$$

where D is a diagonal matrix

$$A^n = M^{-1} D^n M$$

where as D is a diagonal matrix D^n is also diagonal and component ii of D^n is D_{ii}^n. This is nice because if $|D_{ii}| < 1$ for all i then D_{ii}^n tends to 0 as n tends to infinity, so A^n also tends to 0.